MATH110C Homework 2

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1 Exercise 1

Lemma 1.1. Let F be a finite field and n a positive integer. Let K, L be two finite extensions of F of degree n. Then, K and L are F-isomorphic.

Proof. It suffices to show that K and L are the splitting field of the same polynomial by the uniqueness of splitting fields. Let $|K| = |L| = r$. We show this for K, the proof for L is identical.

Notice that the units of K form a multiplicative group of order $r - 1$ and therefore $\forall x \in K^\times : x^{r-1} = 1 \implies x^r = x$. In other words, every element of K is a root of the polynomial $p(x) = x^r - x$. Since p has at most r roots, it follows that every element of K is a root of p with multiplicity 1 and that is all the roots of p. Then, notice that K has to be the splitting field of p since it contains all the roots and **only** the roots of p as its elements. $\square$

2 Exercise 2

Lemma 2.1. Let p be a prime, n be a positive integer and K be a finite field with p^n elements. Let d be a positive integer such that $d \mid n$. Then, K has a unique subfield with p^d elements.

Proof. Let d such that $n = dm$ for some positive integer m. Let $p^d = a$ and $p^n = b$. Using the fact that

$$\Phi_n(x) = \Pi_{d|n}\Phi_d(x)$$

, we can see that $p^d - 1 \mid p^n - 1$. Therefore, K contains the splitting field of $x^a - x$.

Let's now show that roots of this polynomial form a field. They are clearly closed under multiplication and $0, 1$ clearly satisfy the equation. It remains to show that the roots are closed under addition.

Let x, y be roots. Then, $x^a = x$ and $y^a = y$. Then, using the Binomial Theorem and the fact that the characteristic of K is p, $(x + y)^a = x^a + y^a$ and thus $x + y$ is also a root of the polynomial.

Let's now prove uniqueness. If there are two subfields with a elements, every element in both subfield satisfies $x^a - x$, so they have to be the same elements as this polynomial has at most a roots. □

3 Exercise 3

Lemma 3.1. Let $\sigma \in \mathrm{Aut}(\mathbb{R})$. Then, σ is the identity.

Proof. Notice that σ fixes $\mathbb{Q}$. Let's now prove that σ takes positive numbers to positive numbers.

Let $x > 0$. Then, $\exists y \in \mathbb{R} : y^2 = x$. Then, $\sigma(x) = \sigma(y)^2 > 0$. Notice $\sigma(y) \neq 0$ since σ is injective. Then, $\forall x > y \in R : \sigma(x) - \sigma(y) = \sigma(x - y) > 0$, so σ is stricly increasing.

Now, let $x \in \mathbb{R}$. We'll prove that $\forall \epsilon > 0 : |sigma(x) - x| \leq \epsilon$, thus proving that $\sigma(x) = x$. (This lemma is standard in analysis.)

By the density of rationals, there exists $r, q \in \mathbb{Q}$ such that $x - \epsilon < r < x < q < x + \epsilon$. Then, $q = \sigma(q) > \sigma(x) > \sigma(r) = r$, so $|\sigma(x) - x| < \epsilon$. $\square$

4 Exercise 4

Lemma 4.1. The splitting field of $p(x) = x^4 - 9$ is $\mathbb{Q}(\sqrt{3}, i)$.

Proof. Notice that the roots over $\mathbb{C}$ are $\pm\sqrt{3}, \pm i\sqrt{3}$. Therefore, $\mathbb{Q}(\sqrt{3}, i)$ contains all roots of p. Conversely, any field K that contains all four roots contains $\sqrt{3}$ and i since $i = \frac{\sqrt{3}i}{\sqrt{3}}$. $\square$

5 Elman pg.307 Problem 1e

$t^6 + t^3 + 1 = \Phi_9(t)$, so it is irreducible by Problem 7a. Therefore, $[K : \mathbb{Q}] = 6$.

6 Elman pg.307 Problem 2b

Lemma 6.1. Let K be the splitting field of $p(x) = x^6 - 8$ over $\mathbb{Q}$. Then, $[K : \mathbb{Q}] = 4$.

Proof. Note that $\sqrt{2}$ is a solution to p.

Letting, $w = e^{\pi/3i}$, the roots of p are $\sqrt{2}, \sqrt{2}w, sqrt2w^2, \ldots, \sqrt{2}w^5$. Notice that $w = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ using Euler's formula.

Thus, $K = \mathbb{Q}(\sqrt{2}, \sqrt{-3})$. Notice that $\sqrt{-3}$ satisfies $x^2 + 3$ and $\sqrt{-3} \notin \mathbb{Q}(\sqrt{2})$. Thus, $[\mathbb{Q}(\sqrt{2}, \sqrt{-3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{-3}) : \mathbb{Q}(sqrt2)][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2 \times 2 = 4$. $\square$

def A **primitive nth root of unity** is $z \in \mathbb{C}$ s.t. $n := \inf\{k \in \mathbb{N} : z^k = 1\}$.

Let μ_n denote the multiplicative group of nth roots of unity over $\mathbb{Q}$.

Let $\zeta_n := e^{2\pi i/n}$.

Lemma There are $\varphi(n)$ primitive nth roots of unity.

proof Let $f : \mathbb{Z}/n\mathbb{Z} \to \mu_n$ be defined by $a \longmapsto (\zeta_n)^a$.

It's easy to see that f is a group isomorphism.

Now, notice that if $z \in \mathbb{C}$ is a primitive nth root of unity, $w = e^{2\pi i k/n}$ for some k coprime to n. (Otherwise, $\frac{k}{n}$ can be simplified and $z^d = 1$ for some $d < n$.)

Thus, $|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n)$ is the cardinality of the set of primitive roots of unity. ∎

Lemma Let w be a primitive nth root of unity. Then, $\mathbb{Q}(w)$ is the splitting field of $t^n - 1 \in \mathbb{Q}[t]$.

proof Let K be the splitting field of $t^n - 1 \in \mathbb{Q}[t]$.

$\mathbb{Q}(w) \subseteq K$ is immediate.

To see $K \subseteq \mathbb{Q}(w)$, recall that the set of generators for $\mathbb{Z}/n\mathbb{Z}$ is $(\mathbb{Z}/n\mathbb{Z})^\times$.

The isomorphism f from the previous lemma thus finishes the proof. ∎

7 Elman pg.308 Problem 5

Part (a) and (b) were done in Notability. Here's part (c):

Lemma 7.1. Let $\mathbb{Q} \subseteq F$ and $\sigma \in \mathrm{Aut}(F)$. Let $p \in \mathbb{Q}[x]$ be an irreducible polynomial. Then, σ maps roots of p to other roots of p.

Proof. Notice that $\sigma(0) = 0$. Let $\alpha \in F$ such that $p(\alpha) = 0$, i.e. let α be a root of p. Let $p(x) = a_0 + \cdots + a_n x^n$. Then, notice that

$$\sigma(0) = \sigma(a_0 + \cdots + a_n \alpha_n) = a_0 + a_1\sigma(\alpha) + \cdots + a_n\sigma(\alpha)^n$$

Therefore, $\sigma(\alpha)$ is a root of p. □

Lemma 7.2. Let $\omega_1, \ldots, \omega_{\phi(n)}$ be the primitive n-th roots of unity of $t^n - 1 \in \mathbb{Q}[t]$ and let $\sigma \in \mathrm{Aut}(\mathbb{Q}(\omega_1))$. Then, $\sigma(\omega_1) = \omega_i$ for some i such that $1 \leq i \leq \phi(n)$.

Proof. By Exercise 7 (notice I don't use this lemma in Exercise 7, so there's no circular argument), we have that $\Phi_n(x)$ is irreducible. Since ω_i are roots of $\Phi_n(x)$, applying Lemma 7.1 finishes the proof. □

def $\Phi_n(t) := \prod_{i=1}^{\varphi(n)} (t - \omega_i)$

Notice $\Phi_n(t)$ is monic of degree $\varphi(n)$.

Lemma $\forall n \in \mathbb{N}$: $x^n - 1 = \prod_{d|n} \Phi_d(x)$.

proof Clearly, $x^n - 1 = \prod_{\zeta \in \mu_n} (x - \zeta)$.

The formula above just corresponds to partitioning the cyclic group μ_n according to orders of each element. ■

Lemma $\Phi_n(t) \in \mathbb{Z}[t]$.

proof We induct on n. The base case is trivial. Assume the statement holds for all $d < n$.

By the lemma above, $x^n - 1 = f(x)\,\Phi_n(x)$ with $f(x) = \prod_{\substack{d|n \\ d \neq n}} \Phi_d(x)$.

Then, $\Phi_n(x) = \frac{x^n - 1}{f(x)}$ where $x^{n-1} \in \mathbb{Z}[x]$, $f(x) \in \mathbb{Z}[t]$ (this is exactly the inductive assumption). An application of Gauss's Lemma tells us $\Phi_n(x) \in \mathbb{Z}[x]$. ■

8 Elman pg.308 Problem 7a

Lemma 8.1. Let $\omega_1, \ldots, \omega_{\phi(n)}$ be the primitive n-th roots of unity of $t^n - 1 \in \mathbb{Q}[t]$. Let K be the splitting field of $t^n - 1$. Then, for any $1 \leq i, j \leq n$, $\exists \sigma \in \mathrm{Aut}(K) : \sigma(w_i) = w_j$.

Proof. Notice that having any primitive n-th root of unity in the field is enough to generate all the roots. Moreover, any field that has all the roots of unity will also have the primitive roots of unity. Therefore, mapping ω_i to ωj produces a field automorphism that fixes $\mathbb{Q}$ and "recycles" the primitive roots of unity, producing an identical field. □

Let $\Phi_n(x) = p(x)q(x)$ and assume that p is irreducible such that $p(\omega_i) = 0$ for some ω_i.

By contradiction, assume that q is non-constant with some ω_j such that $q(\omega_j) = 0$.

By the lemma above, $\exists \sigma \in \mathrm{Aut}(F) : \sigma(\omega_i) = \omega_j$. By Lemma 7.1, this is a contradiction since ω doesn't map a root of p to another root of p.

Let's now show $\Phi_n(t)$ is irreducible for $n = 3, 4, 6, 8$.

$\Phi_1(t) = t-1$, $\Phi_2(t) = t+1$, $\Phi_3(t) = t^2+t+1$, $\Phi_4(t) = t^2+1$, $\Phi_5(t) = t^4+t^3+t^2+t+1$,
$\Phi_6(t) = t^2-t+1$, $\Phi_8(t) = t^4+1$.

$\Phi_3(t) = t^2+t+1$.

Since $\deg(\Phi_3) \leq 3$, it STS that it doesn't have a root.

By the Rational Root Theorem, the only possible roots are ± 1, and they don't work.

The same argument (word by word) works for Φ_4 and Φ_6.

For $\Phi_8(t) = t^4+1$, note that $t^4+1 = (t^2-\sqrt{2}t+1)(t^2+\sqrt{2}t+1)$.

Since $\mathbb{R}[t]$ is a UFD, $\Phi_8(t)$ can't be factored in $\mathbb{Q}[t]$.

In fact, this is true for all $n \in \mathbb{N}$.